Open Ended Lab

Finite State Machine

Name:	Student ID:

12.1 Objective

- Learn how to implement a finite state machine using Verilog
- Design and implement a simple logic circuit that emulates the blinking lights of a Ford Thunderbird

12.2 Introduction

In this lab, you will design a finite state machine to control the tail lights of a 1965 Ford Thunderbird. There are three lights on each side that operate in sequence to indicate the direction of a turn. Figure 12.1 shows the tail lights and Figure 12.2 shows the flashing sequence for (a) left turns and (b) right turns.

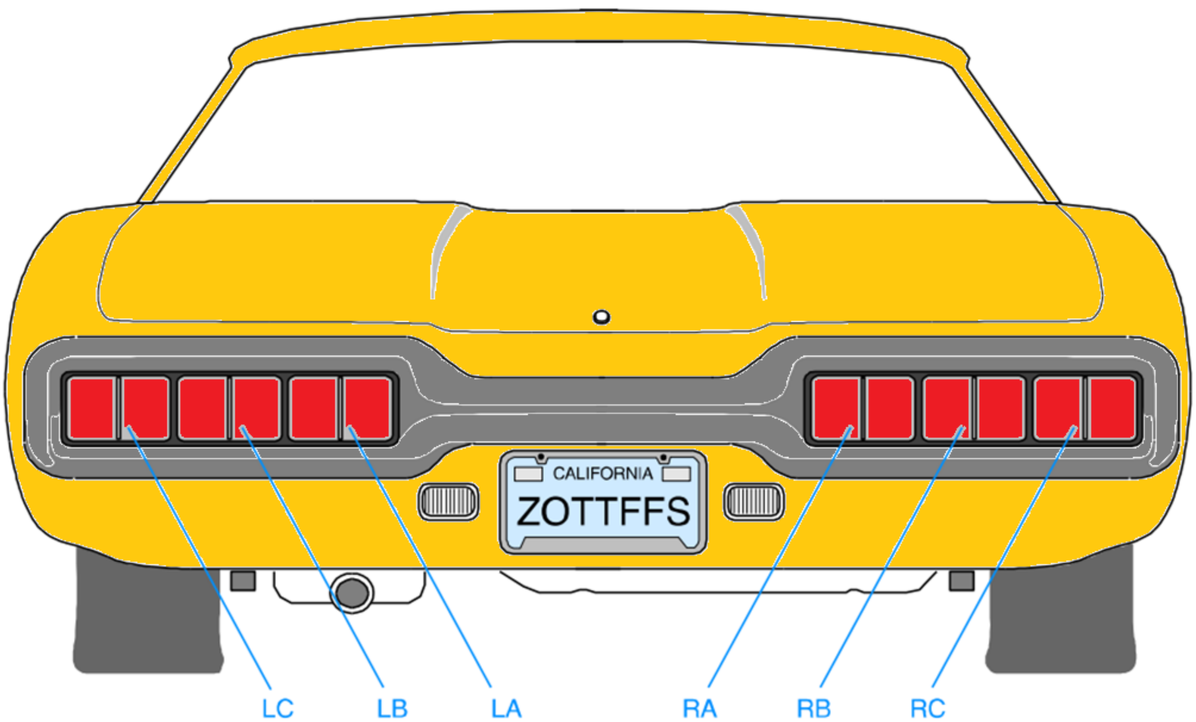

Figure 12.1 - Thunderbird Tail Lights

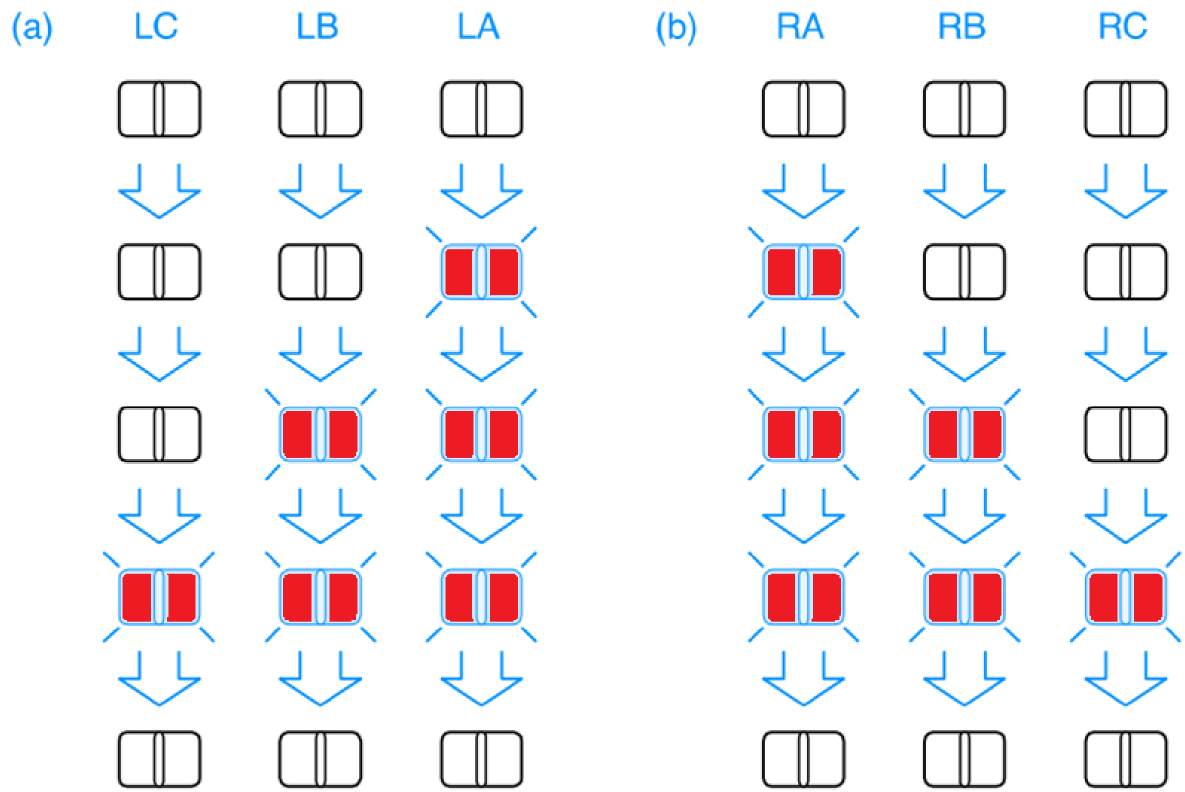

Figure 12.2 - Flashing Sequence

12.3 Design Requirements

Your FSM should take three inputs: **reset**, **left**, and **right** and produces six outputs. Give each state a name and indicate the values of the six outputs LC, LB, LA, RA, RB, and RC in each state. The state diagram should have the following properties:

- On reset, the FSM should enter a state with all lights off (call it **IDLE** state).
- When you press **left**, you should see LA, then LA and LB, then LA, LB, and LC, then finally all lights off again.
- This pattern should occur even if you release **left** during the sequence. If **left** is
  still down when you return to the lights off state, the pattern should repeat.
- The operation when **right** is active should be similar.
- It is up to you to decide on following:

- What to do if the user makes **left** and **right** simultaneously true; make a choice to keep your design *simple*.
- What flipflop to choose for implementation of FSM.
- What binary codes to choose for assigning binary value to state variables.

12.4 Lab Deliverables

i)State Transition Diagram

Sketch the state transition diagram. Indicate the input conditions that cause transitions among states. Use the signals 'left' and 'right' for the direction indicator.

ii)State Assignment

Along with your state transition diagram, pick any state assignment method to map symbolic states to binary values.

iii)State Transition Table

Construct state transition table based on state diagram and chosen state assignment. Add columns for mapping the states to output.

iv)Circuit Diagram of FSM

Obtain input and output equations and sketch circuit diagram to implement the FSM (a sample of Mealy FSM is assuming the D-flip flop as memory element is given in Figure 12.3 as a reference)

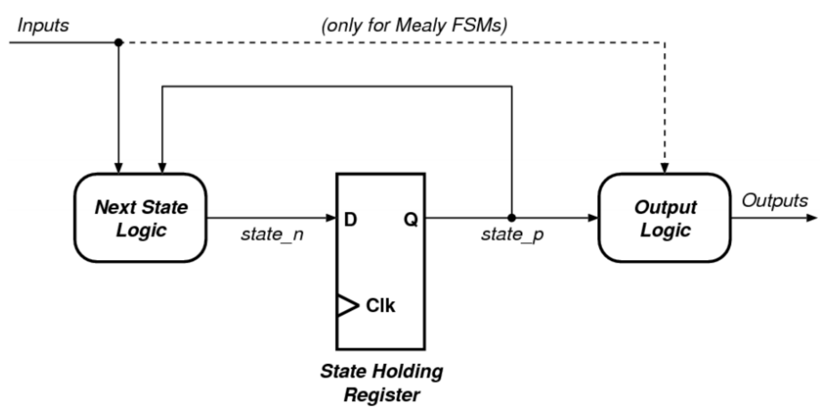

Figure 12.3 - A simple view of an FSM

v)Verilog Implementation

On Eda Playground or Vivado create a new project Add a new Verilog source and implement the finite state machine using what you have learned in the lecture about Verilog. Be sure to include a clock input (clk) and a reset (rst) in your circuit.

vi)Implementing the Clock

The clock available on Basys-3 board is 100 MHz, means that every clock period is 10 nanoseconds long. The entire blinking sequence would be then 40 ns. This is a very short time. light travels less than 250 meters during that time. If we want to see our sequence, we need to find a way to slow down the circuit. This can be achieved in two ways. We can either implement a clock divider circuit that divides the clock by a few million times, or we can generate an enable signal every few million cycles, and then use this enable signal to control our next state transition. Refer to Lab 10 to design a clock divider of your desired value

vii)Top Level Module

Write top module to implement FSM along with clock divider

viii)Define the constraints (to create .xdc file)

Define constrains for your module to implement on Basys-3 Board

ix)Test on BASYS 03 Board

Program Basys-3 board to test and validate your designed modules

Assessment Rubric
Open Ended Lab
Finite State Machine

Name:	Student ID:

Marks Distribution:

	Task No.	OE3 Code	OE5 Results	OE2 Design	OE5 Report
In - Lab	Task i	/10	-		/10
	Task ii	/10	-		
	Task iii	/10	-		
	Task iv	-	-	/10	
	Task v	/10	-		
	Task vi	/10	-		
	Task vii	/05	-		
	Task viii	/05	-		
	Task ix	-	/10		
Total: 100		/60	/10	/10	/10
CLO Mapping		CLO3	CLO3	CLO3	CLO4

Affective Domain Rubric		Points	CLO Mapped
AR 7	Report Submission and Viva	/10	CLO 4

CLO	Total Points	Points Obtained
3	80	
4	20	
Total	**100**	

Open-Ended Lab Evaluation Assessment Rubric

#	Assessment Elements	Level 1: Unsatisfactory	Level 2: Developing	Level 3: Good	Level 4: Exemplary
OE1	**Problem Definition**	Fails to clearly define the problem or research question.	The problem has been defined to some extent but lacks precision or relevance.	The problem or research question has been defined clearly and contextually	The problem has been defined clearly/ Research has been done with precision and relevance.
OE2	**Experimental Design**	The proposed design is unsubstantiated or does not satisfy most of the given constraints. The breakdown of system into features is incomplete and still at lower resolution.	The justification for the proposed design is weak, and it only satisfies some constraints. The breakdown of the system is missing some features and resolution is not appropriate in some places.	The proposed design is substantiated, preferably through proper analysis, and satisfies most given constraints. The breakdown of system into features seems complete, but resolution is not appropriate at some places.	The proposed design is substantiated, preferably through proper analysis, and satisfies all given constraints. The breakdown of system into features seems complete and at appropriate resolution, given students' level.
OE3	**Data Collection**	Measurements are incomplete, inaccurate and imprecise. Observations are incomplete or not included. Symbols, units and significant figures are not included.	Measurements are somewhat inaccurate and imprecise. Observations are incomplete or vague. Major errors are there in the use of symbols, units and significant digits.	Measurements are mostly accurate. Observations are generally complete. Minor errors are present in using symbols, units and significant digits.	Measurements are both accurate and precise. Data collection is systematic. Observations are very thorough and include appropriate symbols, units and significant digits and tasks completed in due time.
OE4	**Analysis & Interpretation**	Fails to analyze data effectively or draw meaningful interpretations.	The analysis of data is incomplete or important trends have been missed or provided limited interpretations.	The data has been analyzed effectively, by identifying evident trends and drawing reasonable interpretations.	In depth analysis has been made by identifying trends, patterns, and relationships. Insightful interpretations have been drawn.
OE5	**Results, Plots and report**	Figures/ graphs / tables are not developed or are poorly constructed with erroneous results. Titles, captions, units are not mentioned. Data is presented in an obscure manner or not presented at all.	Figures, graphs and tables are drawn but contain errors. Titles, captions, units are not accurate. Data presentation is not too clear. Most of the requirements are included in the report but are not well explained.	All figures, graphs, tables are correctly drawn but contain minor errors or some of the details are missing. Report meets all prescribed requirements.	Figures / graphs / tables are correctly drawn, and appropriate titles/captions and proper units are mentioned. Data presentation is systematic. Report meets all requirements, and it is prepared in original and creative way to engage readers.
OE6	**Reflection**	Lacks meaningful self-evaluation or reflection on	Offers limited self-evaluation without	Reflects on the experiment, identifying	Demonstrates critical self-evaluation, identifying

		the experiment.	significant insights.	areas of success and areas for improvement.	strengths, weaknesses, and areas for improvement in the experimental approach.